

Artificial Superintelligence Alliance Vision Paper: CUDOS Edition

Read the original: [Artificial Superintelligence Alliance Vision Paper](#).

D.1. Elevating the ASI Alliance with CUDOS

The integration of [CUDOS](#) into the Artificial Superintelligence Alliance (ASI) marks a transformative step in the pursuit of harnessing scalable, decentralized computing power, an essential component for the development of Artificial General Intelligence (AGI) and Artificial Superintelligence (ASI). This strategic token merger enhances the [Alliance's](#) position as the largest web3 open-sourced, independent entity in AI research and development by complementing recent hardware investments from [SingularityNET](#) and [Fetch.ai](#). By integrating CUDOS, the Alliance meets the critical demand for scalable computing resources, essential for advancing beyond narrow AI towards models that match human cognition. This capability is particularly significant in enabling generalization across domains, achieved through a disruptive AGI technology stack that seamlessly integrates multiple computing paradigms and orchestrates heterogeneous compute nodes within a decentralized network.

CUDOS' cloud infrastructure introduces substantial technical enhancements through its distributed cloud compute model. The architecture inherently reduces the risks associated with bottlenecks and single points of failure, thereby significantly improving overall system robustness and uptime. Moreover, the distributed nature of the infrastructure ensures that sensitive data does not need to be confined to a single location, thereby enhancing security and diversity by minimizing vulnerability to single-site attacks or breaches.

The evolution of AGI systems necessitates infrastructure capable of supporting continuous learning, adaptation, and the integration of vast amounts of data across diverse domains. Decentralized cloud infrastructure, with its inherent flexibility and resilience, provides the ideal foundation for these dynamic knowledge networks. By enabling a fault-tolerant environment where data and compute resources are dynamically allocated in real-time, this infrastructure not only enhances scalability and resilience but also becomes better aligned with the critical need for a foundational layer that supports highly dynamic knowledge networks essential for achieving AGI.

D.2. Strengthening Decentralized Physical Infrastructure: Compute, Storage, and Communication

CUDOS' comprehensive infrastructure and technology stack, which incorporates a robust supply of high-performance cloud compute, will be integral to the distributed systems that form the backbone of the ASI Alliance. This foundational infrastructure underpins the three critical pillars of distributed systems: **compute, storage, and communication**. By integrating CUDOS' decentralized architecture, the ASI Alliance is poised to continue building compound AI systems

that introduce innovative combinations of computing paradigms, enhance the orchestration of computing tasks, and solve previously intractable problems. These advances, in turn, provide actionable economic outputs across a growing range of industries at scale.

The decentralized nature of CUDOS' network extends the Alliance's existing in-house resources across a global network of nodes, thereby enhancing the privacy, resilience, and scalability of its operations. This approach not only alleviates existing compute bottlenecks but also enables the Alliance and its ecosystem to rely on its own sovereign decentralized physical infrastructure. Sovereign infrastructure is crucial for the ASI ecosystem, as it avoids technological capture, censorship, and silicon supply constraints, while ensuring that the infrastructure remains flexible and adaptive to the evolving demands of AGI development.

D.3. Meeting the Exponential Demand for Computational Power

The rapid expansion of AI workloads, driven by increasingly sophisticated models and the necessity for economically viable and trustworthy model-derived inferences, has placed unprecedented demands on computational resources.

CUDOS' token merger into the ASI Alliance directly addresses this challenge by providing on-demand infrastructure that can grow in tandem with the ASI Alliance's expanding needs. A key trend in compute demand is the shift to hybrid neural-symbolic architectures and the progressing AGI technology stack. These paradigmatic shifts in system design address the limitations of traditional generative AI, such as reducing hallucinations that hinder the scaling of task-oriented applications. However, they also require more flexible and composable compute infrastructures to support the near-limitless scale of these advanced AI workloads.

In addition, CUDOS' cloud model offers significant advantages over traditional centralized approaches, including enhanced scalability, cost-efficiency, flexibility, and privacy. For instance, premium AI infrastructure such as NVIDIA H100 GPUs can be accessed on CUDOS at approximately 50% of the cost of Amazon AWS. By enabling access to providers worldwide through CUDOS' distributed cloud architecture, ASI users gain the ability to leverage thousands of the latest AI GPUs, including GB200's, H200's, H100's A100's and MI300s. Each of the suppliers to the compute network operates independently, allowing buyers to access resources globally, thus providing resilient solutions that are not dependent on a single provider.

This enables developers to build virtual nodes and computing resources on the fly to train and fine-tune models, as well as build computationally intensive applications directly on chain and natively through the use of smart contracts.

D.4. Democratization of Compute Resources: A Core Principle

A central pillar of the ASI Alliance's vision is the democratization of access to open infrastructure. CUDOS' decentralized cloud infrastructure is a critical enabler of this vision, providing a secure and scalable environment for hosting mission-critical AI applications. By decentralizing compute resources, the Alliance aims to ensure that these vital tools are not

concentrated in the hands of a few but are instead distributed equitably across a global network. This approach prevents technological lock-in, fosters innovation, and aligns with the ASI Alliance's mission to democratize AI and make its benefits universally accessible.

A key benefit of this integration lies in the empowerment of the ASI community. Through CUDOS' cloud infrastructure, the community gains cost-efficient access to on-demand cloud storage, compute, and communication resources. Users can specify OS images, networking, storage, and compute configurations, enabling the hosting of interfaces, rendering of graphics, training of models, and delivery of inferences. These capabilities are critical for AI development and productization, particularly for products and services designed to serve a global audience where open infrastructure and a high degree of reliability are essential. This represents genuine adoption, allowing a diverse range of entities—regardless of size—to harness the power of advanced AI without the constraints imposed by traditional centralized providers.

The integration of CUDOS into the Alliance's Cloud infrastructure empowers a broad spectrum of developers, enterprises, researchers, and organizations to innovate without limitations. This open-access model is crucial for fostering a vibrant and thriving ecosystem that drives the next wave of AI advancements, ensuring that the benefits of AI are widely distributed and accessible to all.

D.5. ASI Powering the Artificial Superintelligence Alliance

The Artificial Superintelligence (ASI) Alliance heralds a new era in the web3 AI landscape. The founding members welcome CUDOS as the latest strategic partner to merge their token into the alliance. This strategic coalition forms a robust and interoperable ecosystem designed to foster AI innovation at an unprecedented scale. At the core of this ambitious vision is the ASI (\$FET) token—an essential utility token that drives the blockchain-powered, decentralized AI economy forward. The ASI (\$FET) token evolved as a consequence of the token merger of its predecessors FET, OCEAN and AGIX. As it stands, the token is positioned as the foundational utility token of the ASI Alliance and is designed to unify and empower a cohesive ecosystem built around an emerging AI economy. Its utility spans multiple layers, beginning with its role as a layer-1 asset and extending to encompass and enhance the functionalities of the founding protocols.

D.5.1. Layer 1 Utility: Governance, Staking, and Security

Governance: The ASI token empowers holders with a voice in shaping the future of the Alliance. Through on-chain governance mechanisms, token holders can propose and vote on critical decisions, such as protocol upgrades, resource allocation, and practical initiatives. This decentralized governance model ensures that the platform evolves in line with the collective vision of its community, fostering a transparent and accountable environment.

Staking: ASI plays a crucial role in securing the network through a Proof-of-Stake (PoS) model. Token holders can stake ASI to become validators on the Fetch.ai (ASI) mainnet—currently the Fetch.ai mainnet, which will transition to the ASI mainnet in the near future. In return, they earn

rewards while actively contributing to network security and consensus. Despite ASI's interoperability with ecosystems like Ethereum, Cardano, and BNB, staking and validation will be exclusive to the Fetch.ai (ASI) mainnet, ensuring a focused and secure network environment.

Security: The PoS model, underpinned by ASI, ensures robust network security through a decentralized network of validators. This structure not only mitigates the risk of attacks but also promotes long-term stability and trust within the ASI Alliance, making it a resilient platform for decentralized AI and blockchain applications.

D.5.2. Inheriting and Expanding Existing Utilities

The ASI token is not just about creating new utilities; it also integrates and amplifies the strengths of the founding and partner protocols, ensuring a seamless transition and enhanced functionality:

Ocean Protocol - Data Farming: ASI expands upon Ocean Protocol's data monetization framework. Token holders will be able to stake ASI to create data pools, enabling users to access valuable datasets while data providers earn rewards. This integration allows ASI to tap into the growing market for decentralized data services, further extending its utility.

SingularityNET - AI Marketplace Access: ASI becomes the native token for accessing a wide range of AI services and products. This integration streamlines AI-mediated transactions, reduces friction, and broadens the ASI token's use case, making it a critical component in the decentralized AI economy.

Fetch.ai - Agent Search and Discovery: The ASI token powers the agent registration and discovery mechanisms within the Fetch.ai (ASI) network. By ensuring a smooth transition for existing users, ASI enhances the discoverability of AI agents and solidifies its role within the agent ecosystem.

D.5.3. CUDOS Integration: Unlocking New Horizons of Utility

The token merger with CUDOS introduces a transformative new layer of utility into the ASI tokenomics framework:

Cost-Effective Compute Power: CUDOS brings a decentralized cloud computing network into the ASI Alliance, enabling ASI token holders to access and pay for computational resources at competitive rates. This capability is crucial for the development and deployment of sophisticated, resource-intensive AI applications, democratizing access to high-performance computing power.

Expanding the Compute Marketplace: The integration of ASI with CUDOS fosters the growth of a dynamic compute marketplace within the Alliance. Developers, enterprises, and innovators

can leverage ASI to both supply and access affordable, scalable computing power, driving further innovation and expanding the ecosystem's utility.

Staking for reduced Compute costs: Developers will be able to stake ASI to reduce their compute costs even further.

D.5.4. A Gateway to the Future of AI

The ASI (\$FET) token transcends the traditional role of a currency; it is the linchpin of an emerging economy where the creation, deployment and public use of AI-centric products and services are both funded and paid for through the blockchain. Through its multi-layered utility, the ASI token powers every aspect of the ASI Alliance, from governance and security to AI agent interactions, data monetization, and access to decentralized computing resources. As the Alliance grows and the technology stack evolves, the ASI token will be central to emerging use cases, positioning both community members and developers at the forefront of creating, deploying, and monetizing next-generation AI technologies. This includes the development of compound AI systems and, eventually, a framework for artificial general intelligence (AGI) and superintelligence, cementing ASI's place as a cornerstone of the decentralized AI economy.

D.6. Strategic Impacts and Future Directions

The inclusion of CUDOS in the Alliance represents a significant milestone, greatly enhancing the Alliance's capacity to scale compute resources in support of AGI and ASI development. This collaboration aligns seamlessly with the broader objective of advancing beyond narrow AI and toward general and superintelligence, while introducing the strategic advantage of sovereign compute—computational resources owned and controlled by the suppliers to the compute network, free from reliance on external providers.

Sovereign compute is essential as it grants the ASI Alliance full autonomy over its computational infrastructure, reducing dependency on centralized third-party cloud providers and significantly enhancing the resilience of the ecosystem. This autonomy ensures that mission-critical AI applications can operate with greater independence, safeguarding them against risks associated with external control or complete service disruptions.

Moreover, CUDOS' integration leverages blockchain technology and cryptographic primitives, embedding security, transparency, and trust at the core of the ASI ecosystem. This robust technical foundation is crucial for maintaining the integrity of compute resources, especially as the Alliance scales its efforts in AGI and ASI development. The use of decentralized, secure systems underpins the Alliance's commitment to a transparent and trustworthy AI infrastructure, essential for the development of technologies that will shape the future.

The trade-off between latency and resilience is another critical consideration within this decentralized model. Users can decide which global region to deploy their computing resources into to reduce the physical distances and therefore the latency between the nodes or from the nodes to the end users of the resources. Additionally, the resilience offered by a decentralized

system is unmatched. Resilience, driven by the very nature of distributed systems, enables the deployment of a growing range of highly impactful, commercially viable products by ensuring that the system can withstand single supplier and location failures or attacks on individual nodes without compromising the overall functionality of the network.

The addition of CUDOS brings with it significant skills and advisors to the Alliance. The founding team of CUDOS previously founded and built the UK's largest data center and cloud network. Plus a core advisory team including the Director for AI for AMD and the 2 prior presidents of Playstation, with Playstation being one of the world's largest distributed device networks in the world. It also brings CUDOS' NVIDIA Cloud Partnership which is key to maintaining a leading edge in the market of NVIDIA AI infrastructure.

The token merger with CUDOS will naturally lead to improvements and extensions to the underlying CUDOS framework, moving it further toward rich and robust decentralization of all aspects of AI compute. As one part of this, CUDOS will be seamlessly integrated with the NuNet protocol and tooling, creating a more robust and versatile distributed computing ecosystem. NuNet, a SingularityNET spin-off project, specializes in decentralized computing by enabling the monetization of idle CPU and GPU resources across edge devices. This allows smaller contributors to participate in the network by offering their unused computing power. In contrast, CUDOS focuses on data center-centric solutions, empowering larger contributors to deploy AWS-like data centers and instantly connect to a growing compute marketplace. This collaboration enhances the Alliance's compute capabilities, empowering both the core team and the broader community to utilize and monetize compute resources optimized for geographic proximity. CUDOS' infrastructure will support applications requiring continuous, high-demand compute power, while NuNet will bridge the gap between geographic location and compute demand, ensuring reliable and low-latency computing near the point of need. This reduction in geographic latency strengthens the Alliance's ability to deliver a dependable compute marketplace, both locally and globally.

Furthermore, the rising demand for trustworthy AI inferences underscores the need for a robust decentralized technology stack, which includes an autonomous containerized (e.g. Dockerized) compute deployment and orchestration layer. In this context, CUDOS and NuNet are collaborating to significantly advance decentralized container orchestration technology over the next 12 months. Containerization and virtualization, a well-established practice in the DevOps industry, involves sandboxing and isolating processes, ensuring AI tasks run independently without interference. This maximizes performance, security, resource efficiency, and the ease of deploying autonomous services. In this context NuNet is developing a framework for the distribution and orchestration of AI and other compute workloads across the network, matching and leveraging the best available hardware for a job while CUDOS provides its distributed infrastructure at the bare metal and virtualization layers. This layered architecture—bare metal as the foundation, virtual machines offering flexible compute environments, and decentralized containers on top—ensures optimal resource allocation and performance for AI processes. From a practical standpoint, both the core team, the more proximal ecosystem and the devops industry as a whole benefit significantly. By first developing and subsequently integrating

decentralized containers into the existing technology stack, the Alliance satiates the significant demand on CUDOS' part for serverless computing, whilst extending the utility of NuNet. One of the first demand vectors for serverless computing is distributed AI inference to scale out generative AI workloads analogous to the work of Stable Diffusion. Moreover the Alliance strengthens its position as a provider of an increasingly competitive AI development software stack.

Consequently, following the token merger, CUDOS will support the deployment of computing in clusters with very low inter-machine latency, ideal for both large and small training workloads. Additionally, the ability to deploy bare metal infrastructure, which means direct access to computing hardware which is key to high-performance computing infrastructure for AI because it provides the fastest performance possible which is critical when training AI. This capability enables the full potential of training and inference across a global network of compute resources, powering the next generation of AGI and ASI. Together this provides the full infrastructure stack for AI workloads.

D.7. Tokenomics

D.7.1 Token Conversion Ratios

The conversion ratios between \$CUDOS and \$FET (ASI) tokens are set at:

- **Token Merger Rate:** CUDOS 112.427:1 FET - \$CUDOS tokens migrate to \$FET, at this conversion rate.
- **Token Merge Fee:** 5%
- **Conversion Ratio:** CUDOS 118.344:1 FET calculated as a 5% adjusted rate (112.427 / 95%) -> if a user has 119 CUDOS this would be converted into ~1 FET
- **Average Price:** 30-Day Window (\$0.0084497923197215x)
- **Vesting (public):** 3 Months
- **Vesting (treasury):** 10 Months
- **Supply Increase:** 88,946,755.672 FET (i.e. 10,000,000,000 / 112.427)

As a non-founding member of the Alliance, CUDOS have agreed that all token merge transactions be vested between 3-10 months, and include a 5% transaction fee. This fee is provided to the ASI Alliance for marketing, research and development, integration and legal fees, and any other costs associated with the token merger. This fee is pre-calculated within the conversion ratio.

The price of each token was fixed before the announcement of the merger to avoid the valuation being impacted by sudden post-announcement market activity.

D.7.2 Token Conversion Process

The mechanical process of executing the token merger is planned as follows.

- **Ethereum**
 - CUDOS to FET migration contracts will be deployed at <https://www.singularitydao.ai/migrate-asi>
- **CUDOS Chain / Fetch (ASI) Chain**
 - A chain merge will be executed, in which existing CUDOS Chain tokens will be automatically migrated to Fetch (ASI) Chain and CUDOS Chain validators will be invited to join the Fetch.ai (ASI) mainnet..

D.8. Conclusion

The integration of CUDOS into the ASI Alliance is a pivotal development that elevates the compute strategy, ensuring that the necessary infrastructure is in place to support the next generation of AI technologies. This collaboration significantly strengthens the Alliance's position as the largest web3 open-sourced, independent entity in AI research and development. By adding CUDOS' expertise in distributed AI computing, we will significantly accelerate the advancement of decentralized Artificial General Intelligence (AGI) and, ultimately, Artificial Superintelligence (ASI).

This token merger not only bolsters our technological capabilities but also amplifies our commitment to driving the future of AI in a transparent, open, and inclusive manner. The benefits extend far beyond the Alliance, empowering a global community of developers, researchers, and organizations to innovate without constraints. Together, we are laying the foundation for a future where the transformative potential of AI is accessible to all, ensuring that the advancements we achieve are widely distributed and universally beneficial. The Alliance invites the broader community to join us in this journey, as we continue to shape the future of AI with a vision that is as meaningful as it is ambitious.